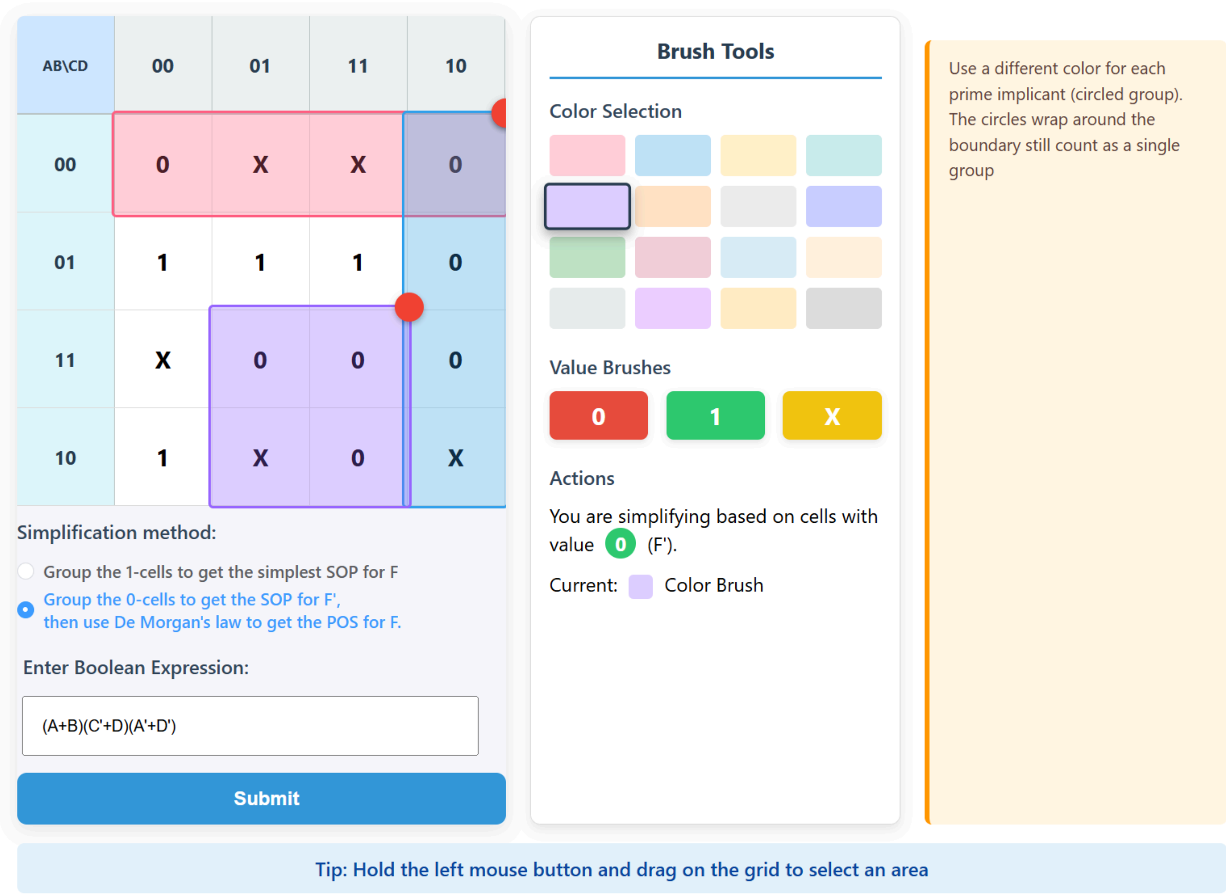
Digital Logic - Assignment I

1.Simplify the following four-variable Boolean functions to simplest POS form using K-map method: $F(A,B,C,D) = \Pi(0, 2, 6, 11, 13, 14, 15) + d(1, 3, 9, 10, 12)$.



2.Obtain the simplified Boolean expressions for output F1 and F2 in terms of the input variables in the following circuit.

F1

1 = $((AB)'((A+B)')')\oplus(C')'$ ×

2 = $(A'+B')(A+B)\oplus C$ ×

3 = $(A'B+AB')\oplus C$ ×

4 = $C(A'B+AB')'+C'(A'B+AB')$ ×

5 = $C(A+B')(A'+B)+A'BC'+AB'C'$ ×

6 = $ABC+A'B'C+A'BC'+AB'C'$ ×

7 = $C(AB+A'B')+C'(A'B+AB')$ ×

Add simplification step (or press Enter)

Please add ' after subscript, e.g., Qt+1'

Allowed Variables:

A B C

Simplify the following expression:

F2

1 = ((AB)'C'+(A+B)')' ×

2 = (AB+C)(A+B) ×

3 = AB+AC+BC ×

4 = A(B+C)+BC ×

Add simplification step (or press Enter)

Please add ' after subscript, e.g., Qt+1'

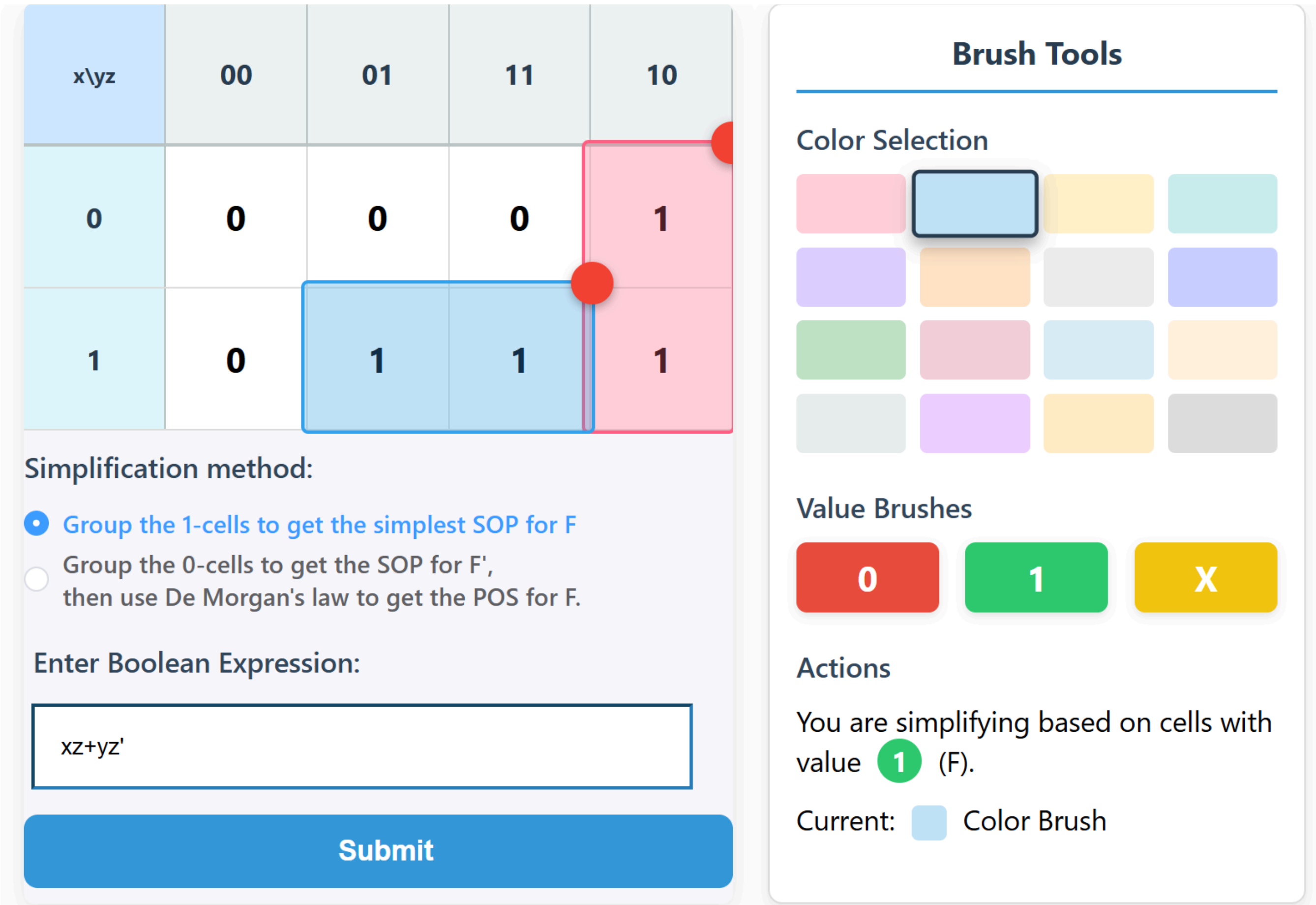
Allowed Variables:

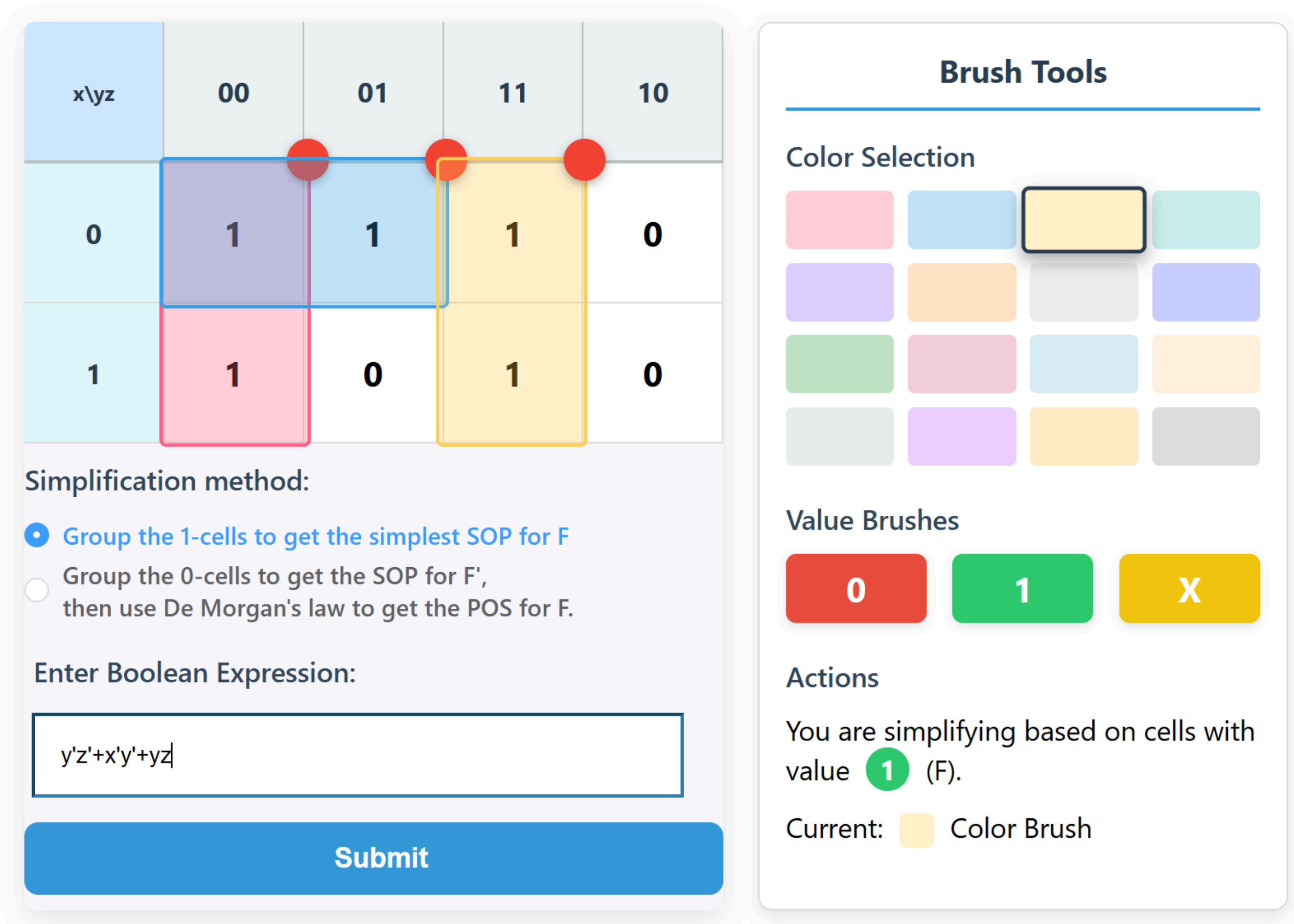
A B C

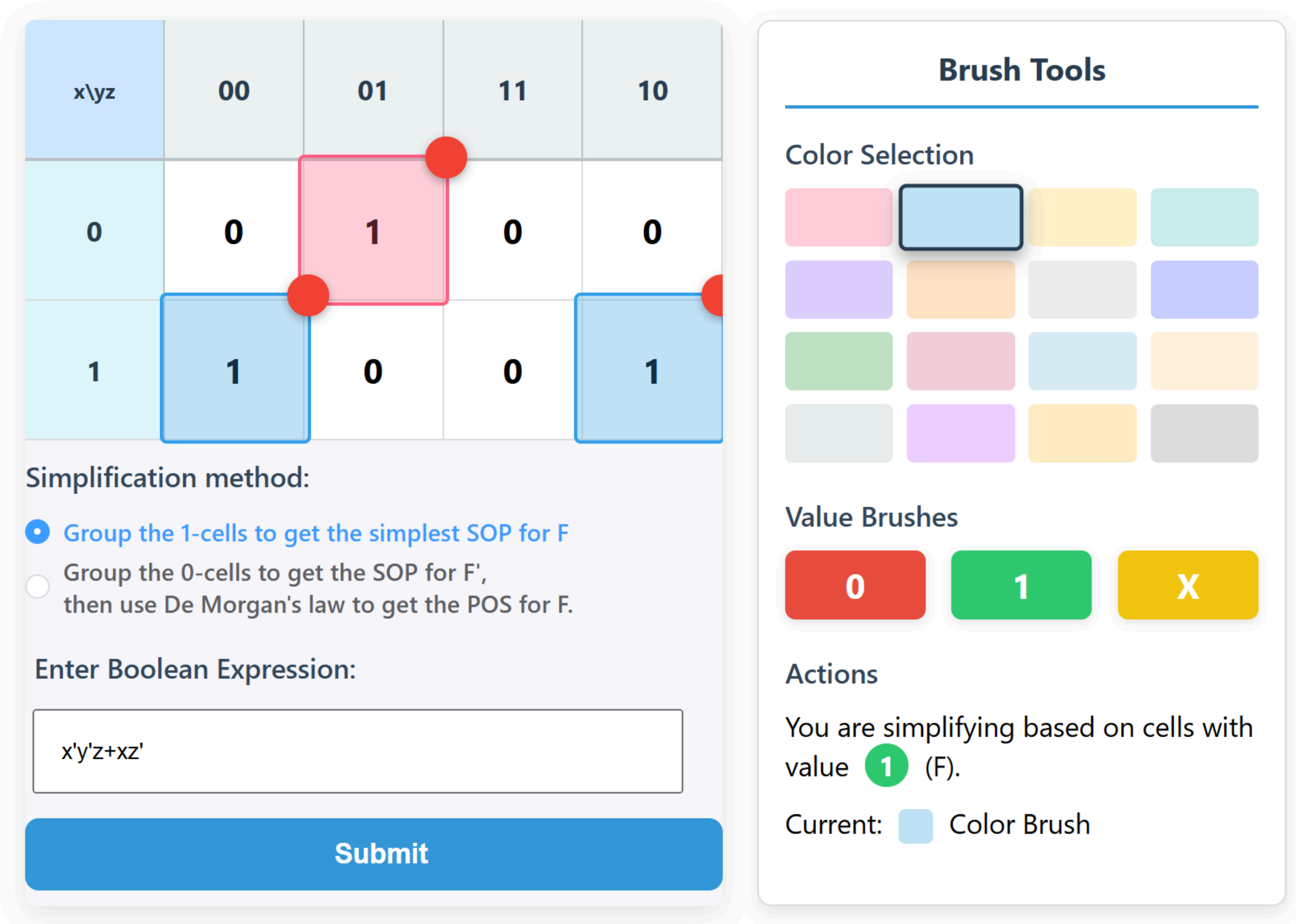
A	B	C	T1	T2	T3	T4	F1	F2
0	0	0	1	1	1	0	0	0
0	0	1	1	1	0	0	1	0
0	1	0	1	0	1	1	1	0
0	1	1	1	0	0	1	0	1
1	0	0	1	0	1	1	1	0
1	0	1	1	0	0	1	0	1
1	1	0	0	0	0	0	0	1
1	1	1	0	0	0	0	1	1

3.Design a combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, or 2, the binary output is two greater than the input. When the binary input is 3, 4, 5, 6, or 7, the binary output is one less than the input. You need to provide:

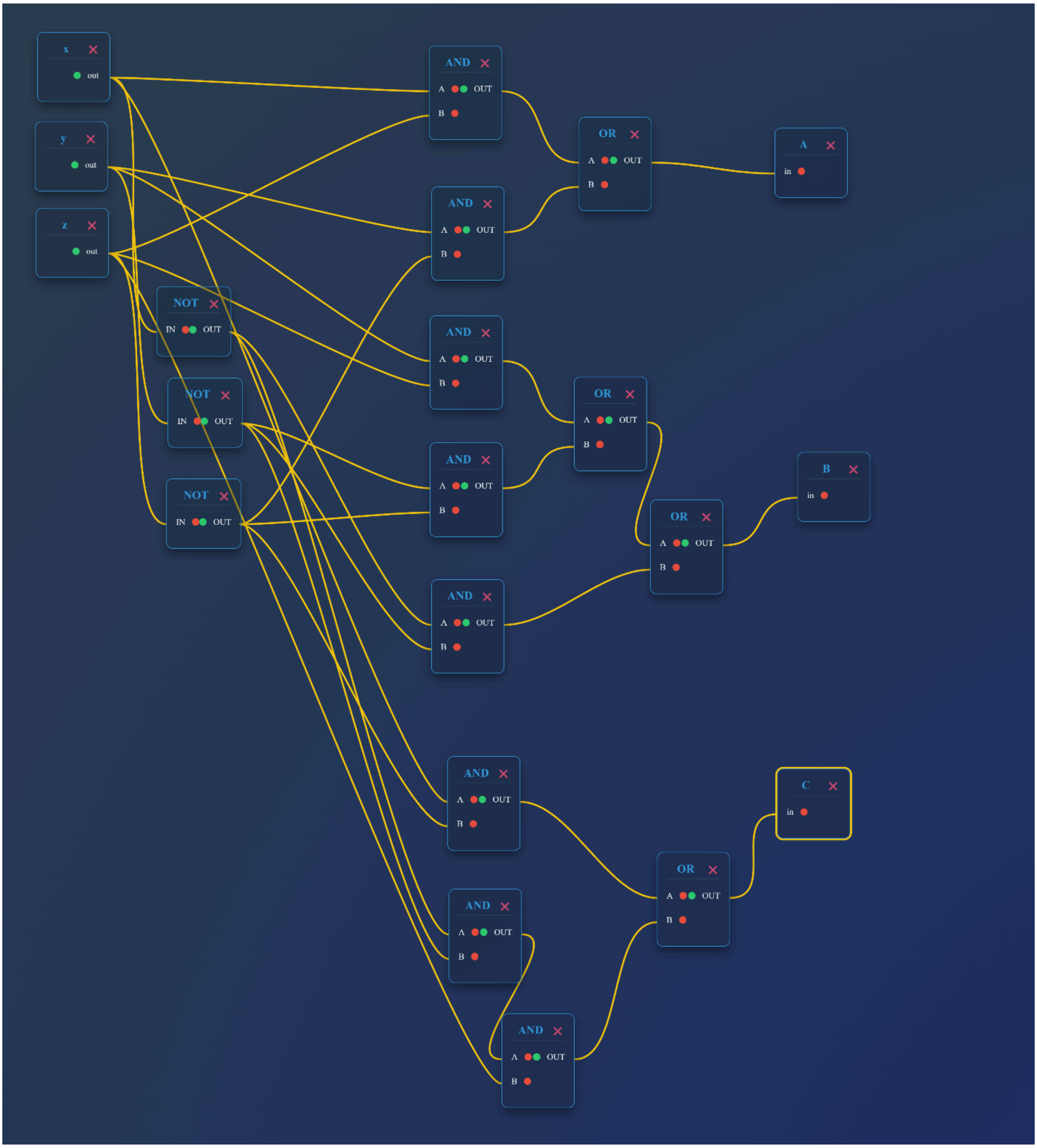
x	y	z	A	B	C
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0


x\yz
00
01
11
10
0
0
0
0
1
1
0
1
1
1
Simplification method:
Group the 1-cells to get the simplest SOP for F
Group the 0-cells to get the SOP for F',
then use De Morgan's law to get the POS for F.
Enter Boolean Expression:
xz+yz'
Submit
Brush Tools
Color Selection
Value Brushes
0
1
X
Actions
You are simplifying based on cells with value 1 (F).
Current: Color Brush

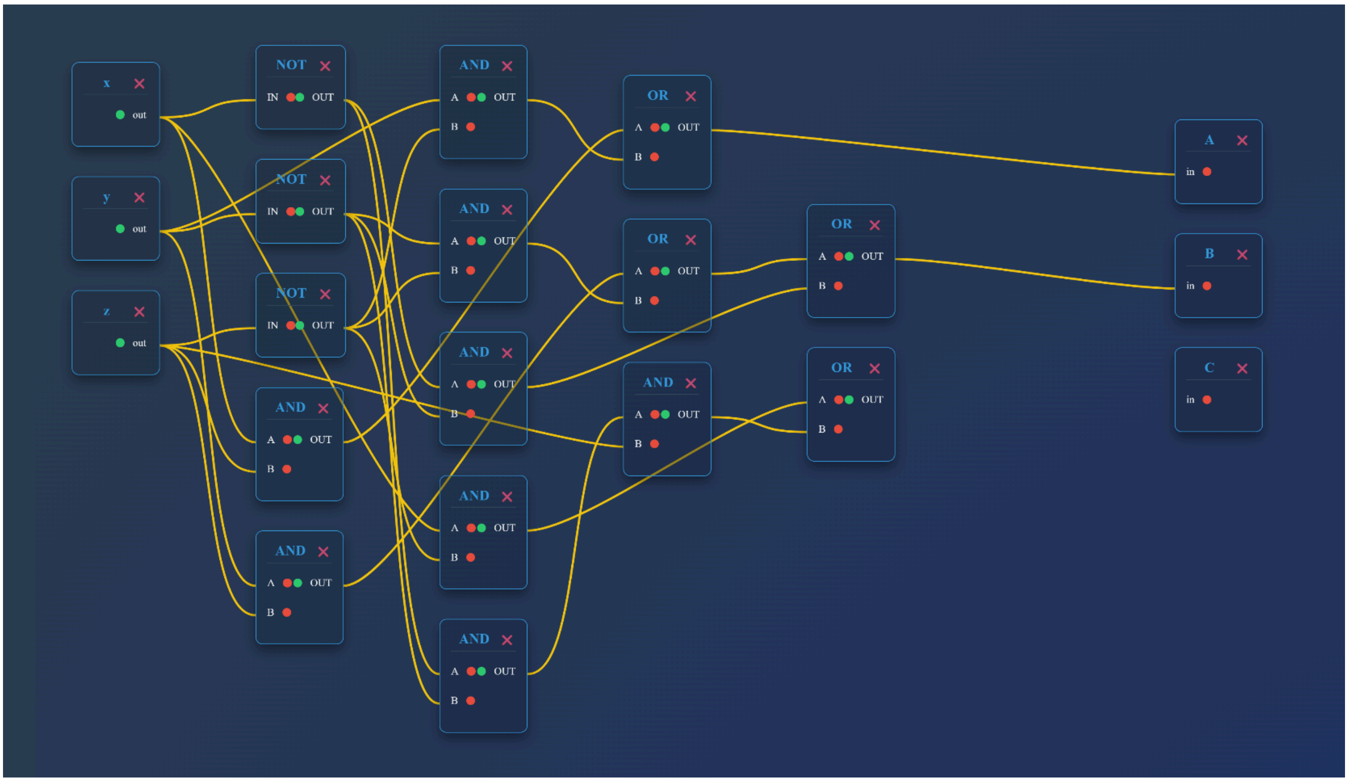

x\yz
00
01
11
10
0
1
1
1
0
1
1
0
1
0
Simplification method:
Group the 1-cells to get the simplest SOP for F
Group the 0-cells to get the SOP for F',
then use De Morgan's law to get the POS for F.
Enter Boolean Expression:
y'z'+x'y'+yz
Submit
Brush Tools
Color Selection
Value Brushes
0
1
X
Actions
You are simplifying based on cells with value 1 (F).
Current: Color Brush



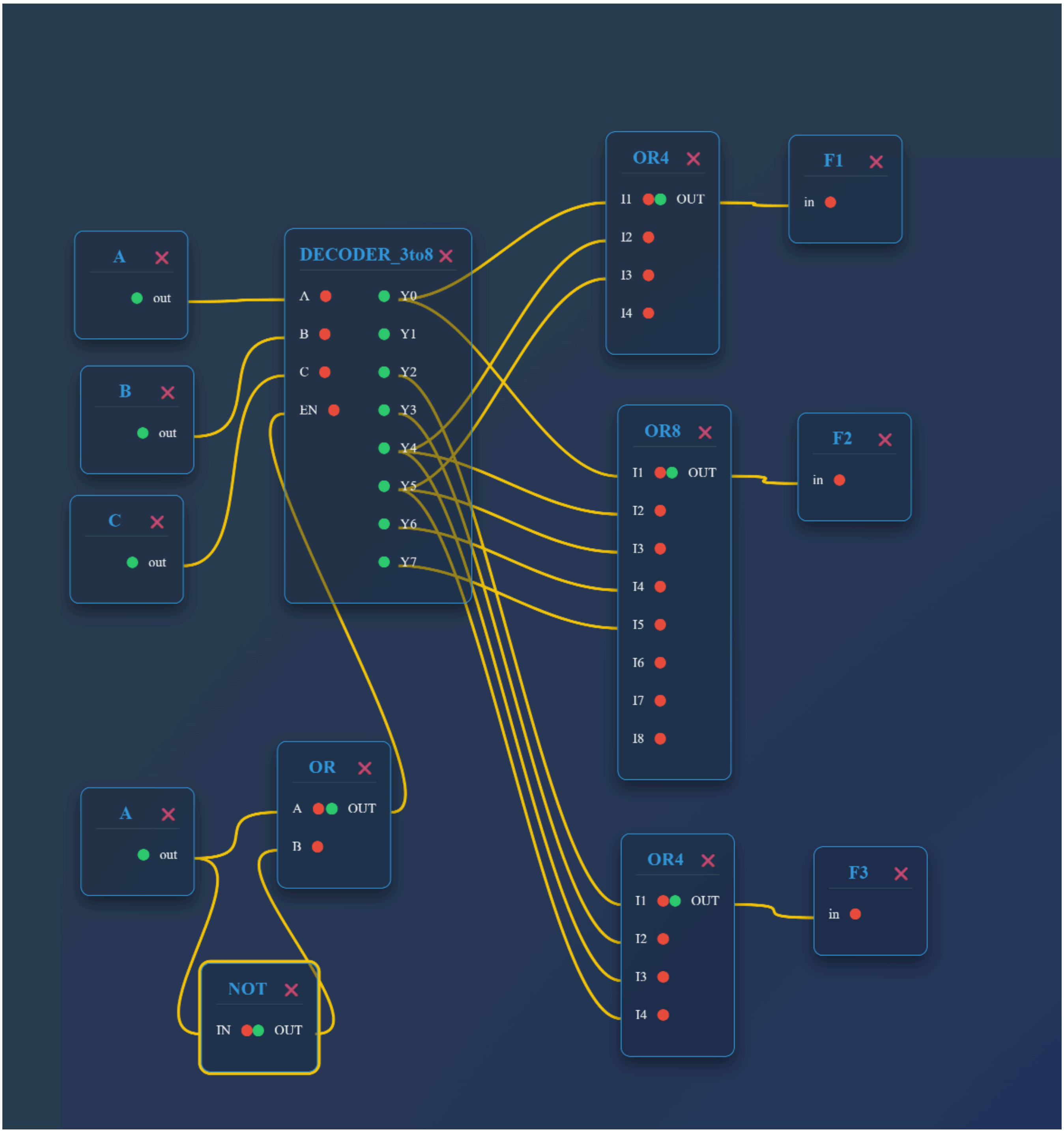
线很乱可能难以辨认，建议直接对着 expression 自己画一下，很简单。



x
out
y
out
z
out
NOT
IN
OUT
AND
A
OUT
B
OR
A
in
B
C


x
out
y
out
z
out
NOT
IN
OUT
NOT
IN
OUT
NOT
IN
OUT
AND
A
OUT
B
AND
A
OUT
B
AND
A
OUT
B
AND
A
OUT
B
AND
A
OUT
B
AND
A
OUT
B
AND
A
OUT
B
OR
A
OUT
B
OR
A
OUT
B
AND
A
OUT
B
OR
A
OUT
B
OR
A
OUT
B
A
in
B
in
C
in

4.A combinational circuit is defined by the equations F1=AB+A’B’C’ F2=A+B+C’ F3=A’B+AB’ Design a circuit which implements these three equations using a decoder and OR gates external to the decoder, and draw the logic diagram.



5.An 8:1 multiplexer has inputs A, B, and C connected to the selection inputs S2, S1, and S0, respectively. The data inputs I0 through I7 are as follows: I1=I2=0; I3=I5=I7=1; I0=I4=D; and I6=D’. Determine the Boolean function F(A, B, C, D) that the multiplexer implements. You need to:

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1

1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

AB\CD	00	01	11	10
00	0	1	0	0
01	0	0	1	1
11	1	0	1	1
10	0	1	1	1

Simplification method:

- Group the 1-cells to get the simplest SOP for F
- Group the 0-cells to get the SOP for F', then use De Morgan's law to get the POS for F.

Enter Boolean Expression:

AC+BC+B'C'D+ABD'

Submit

Brush Tools

Color Selection

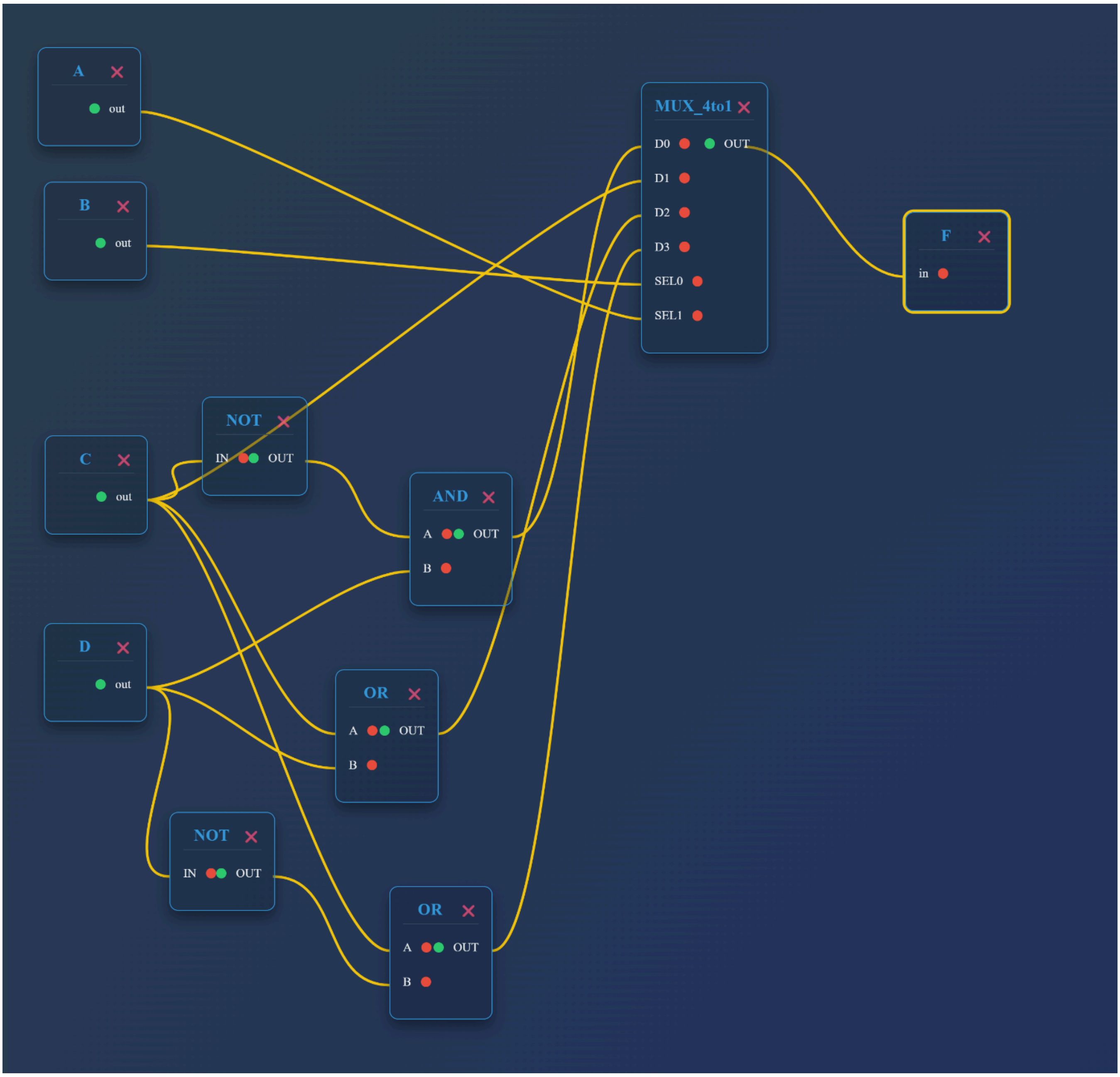
Value Brushes

0 1 X

Actions

You are simplifying based on cells with value 1 (F).

Current: Color Brush


A
out
B
out
C
out
D
out
NOT
IN
OUT
NOT
IN
OUT
AND
A
OUT
B
OR
A
OUT
B
OR
A
OUT
B
MUX_4to1
D0
OUT
D1
D2
D3
SEL0
SEL1
F
in

6.Construct a 16x1 multiplexer with two 8x1 and one 4x1 multiplexers.